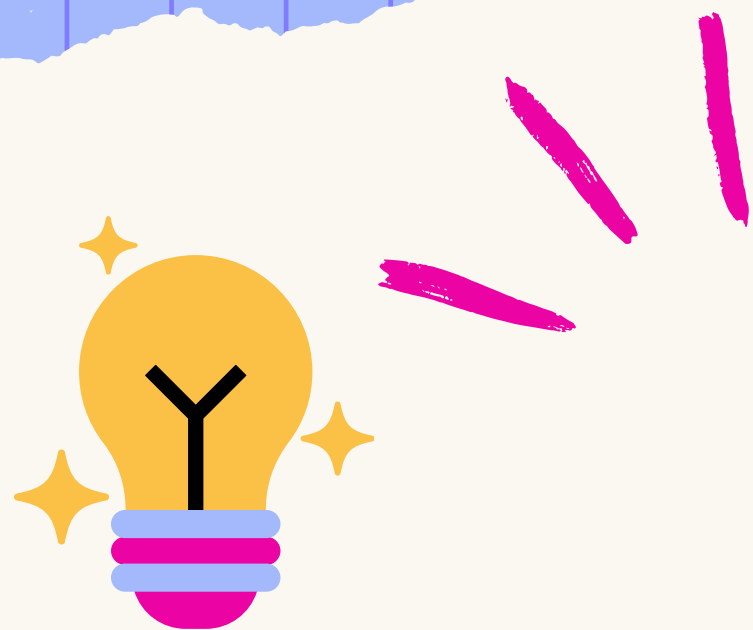




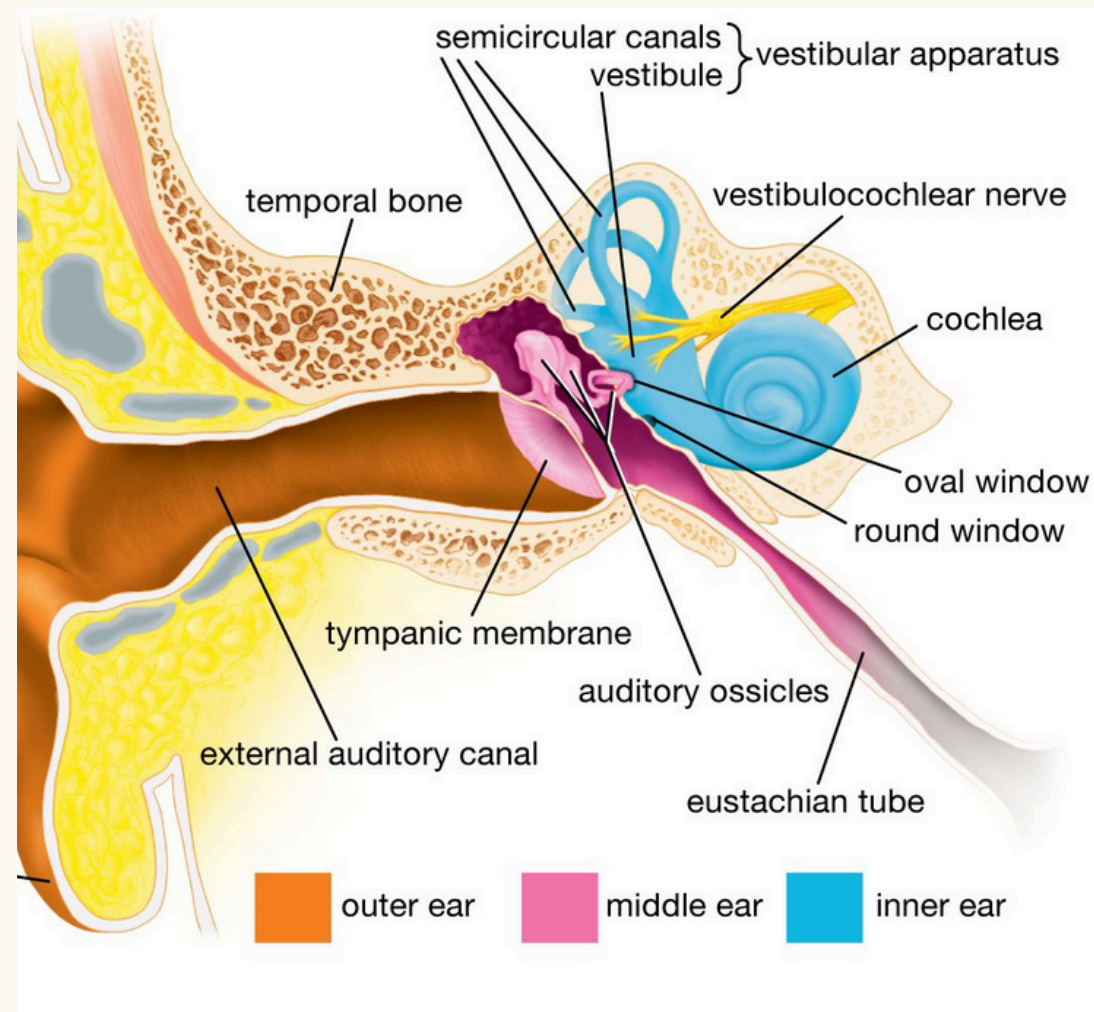
Presentation on

DESIGN AND ANALYSIS OF A TOTAL OSSICULAR REPLACEMENT PROSTHESIS (TORP)

BY: SANDEEP ROY,
ROLL NO. 230710102098

GUIDE: DR. APURBA DAS, IIEST SHIBPUR
DEPARTMENT OF MECHANICAL ENGINEERING,

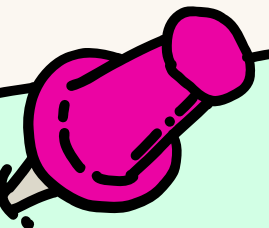
Introduction



- Conductive hearing loss → caused by damaged ossicles
- TORP → surgical implant to restore sound transmission
- Requirements:
 1. Biocompatible material
 2. Strong yet lightweight
 3. No resonance in 100–6000 Hz range

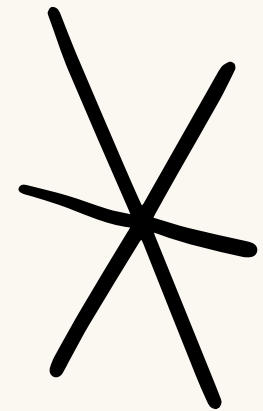
“Why this matters: restoring hearing & quality of life”



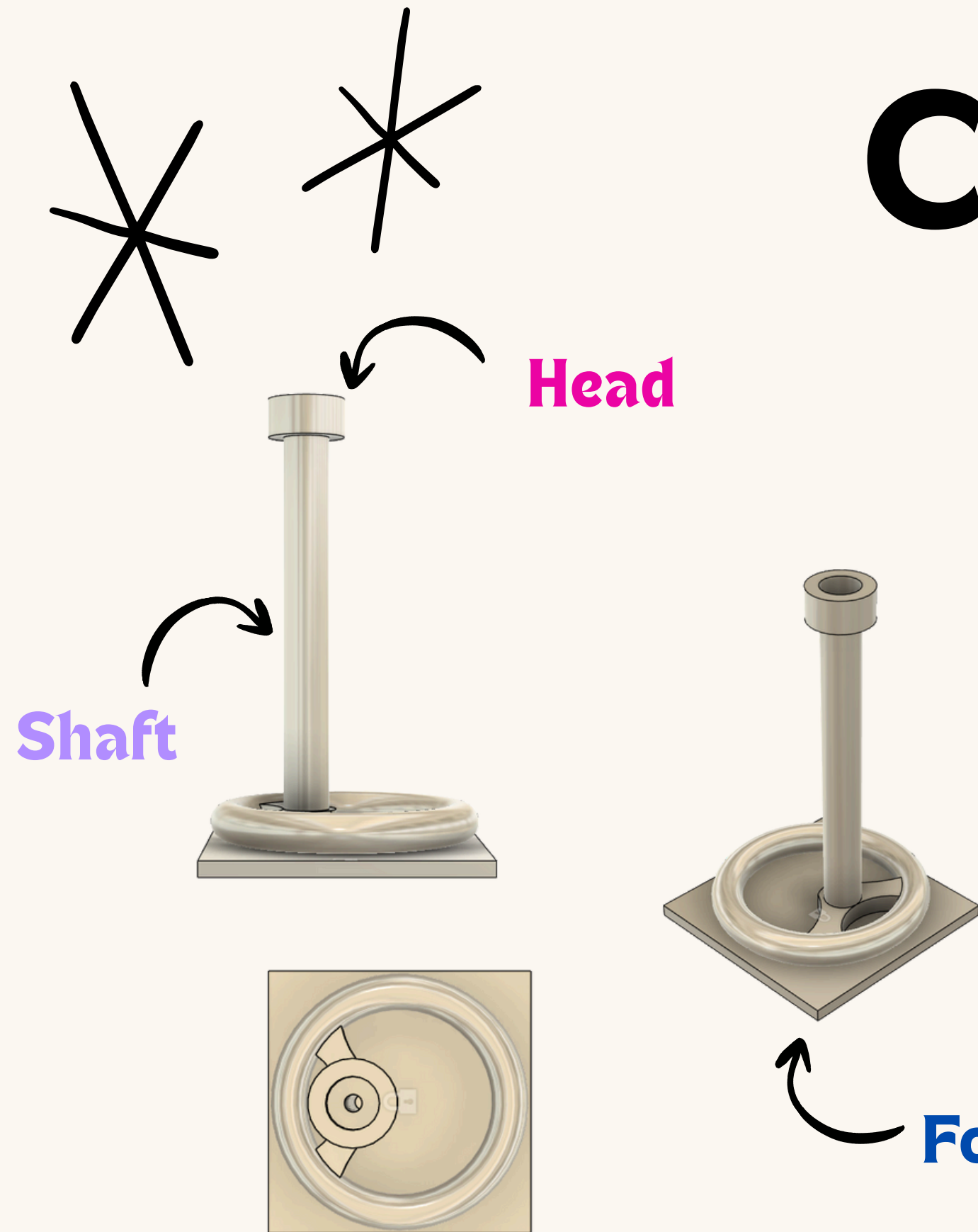


Objectives

1. Build CAD model of TORP (Fusion 360)
2. Select implant-grade material (Ti-6Al-4V)
3. Perform static stress analysis under acoustic load
4. Perform modal frequency analysis
5. Evaluate mechanical safety & reliability



CAD Design

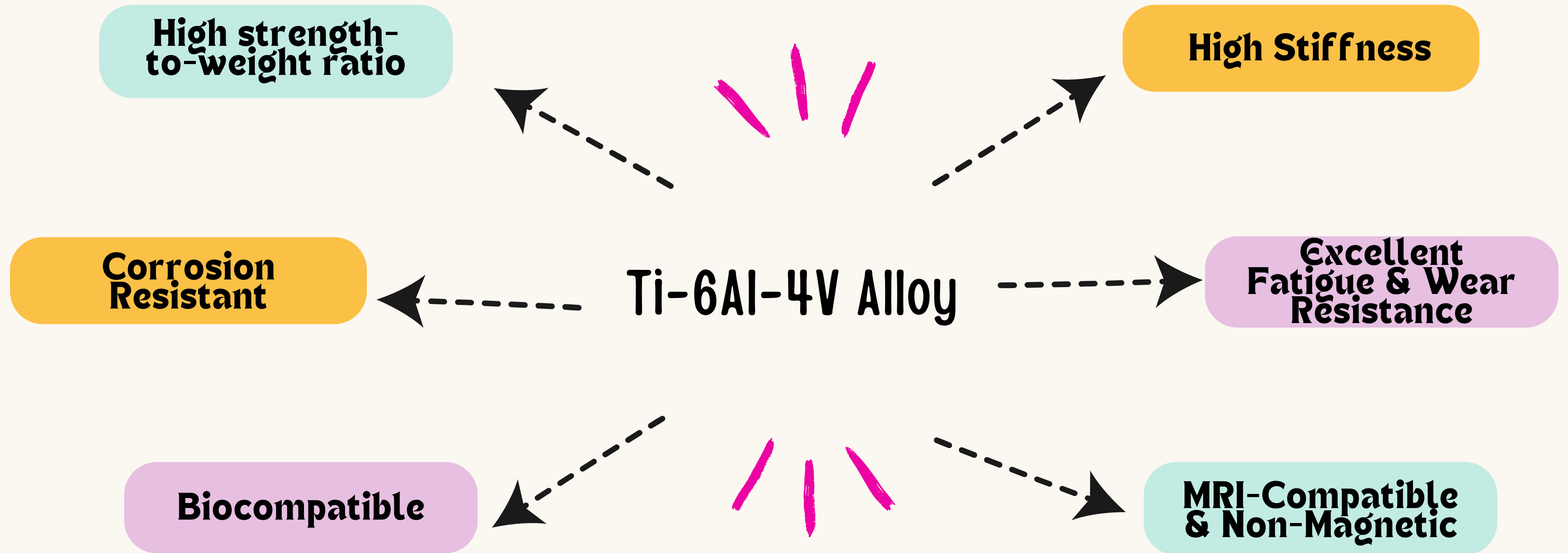


MAIN PARTS:

1. Footplate – stable contact with stapes
2. Shaft – connects base to head
3. Head – contact with the tympanic membrane

Key Features: Compact | Lightweight | Smooth transitions | Manufacturable

Material Selection



► Simulation Methodology

Static Stress Analysis

Load = 5.0×10^{-5} N

Pressure = 1.0×10^{-6} MPa

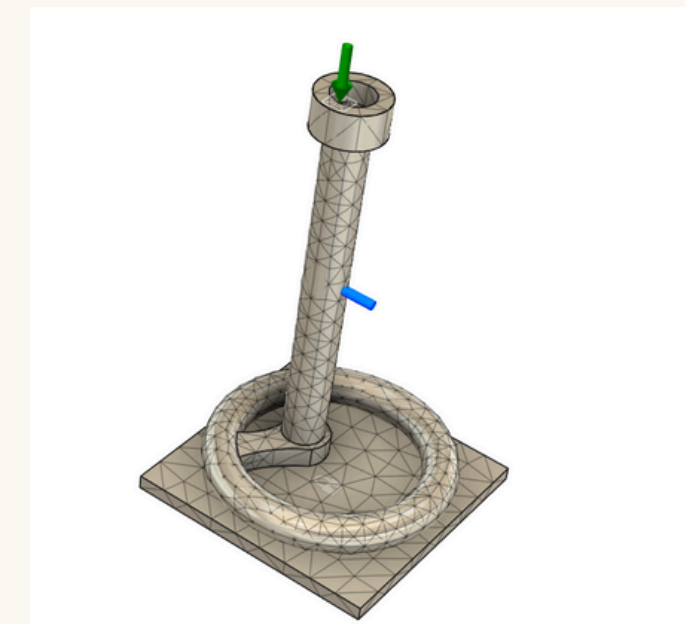
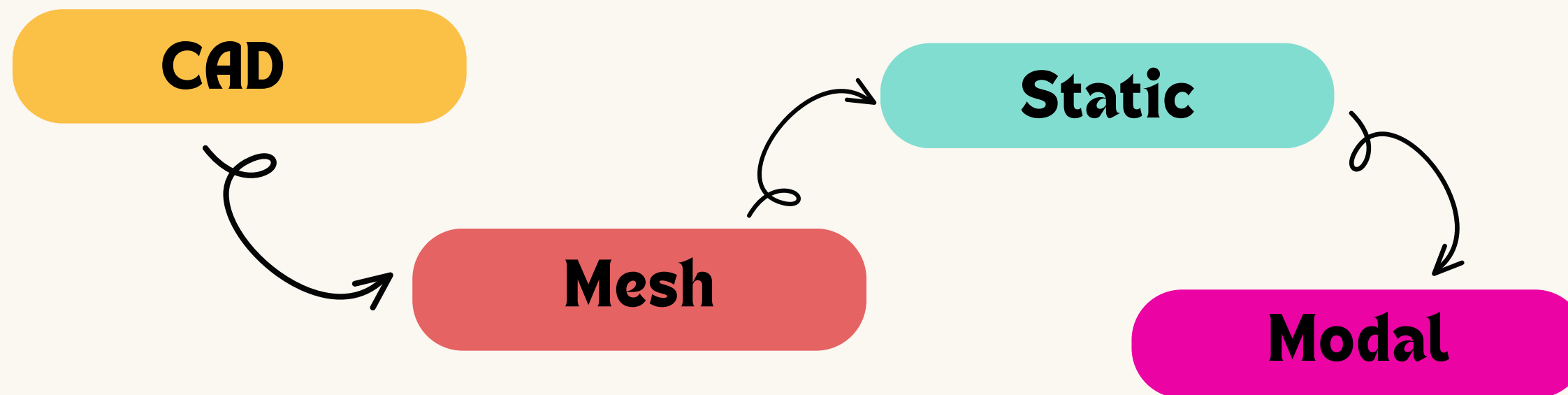
Footplate fixed

Modal Analysis

8 eigenmodes

Goal: Avoid resonance in
100–6000 Hz

Flowchart:

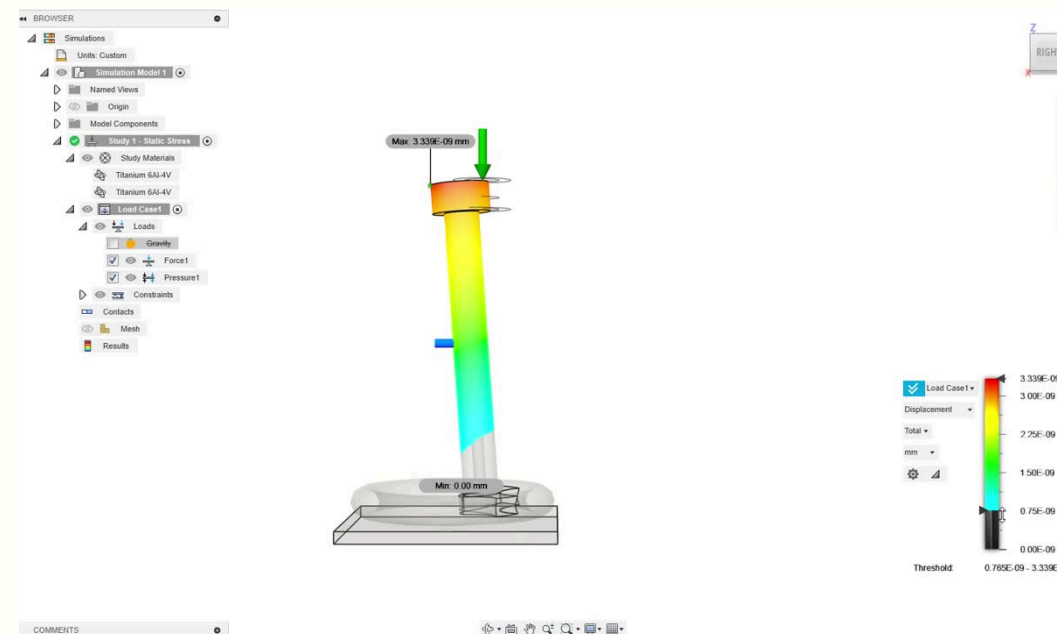


Static Analysis Results

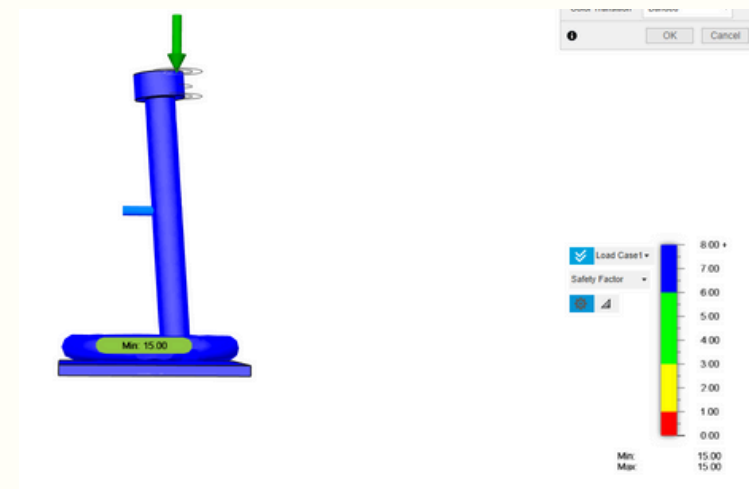
- 1 Max displacement: 3.339×10^{-9} mm
→ negligible
- 2 Max stress: 3.378×10^{-5} MPa
<< yield strength
- 3 Safety factor: 15.0 → very safe
- 4 Prosthesis behaves like rigid body



Stress distribution plot



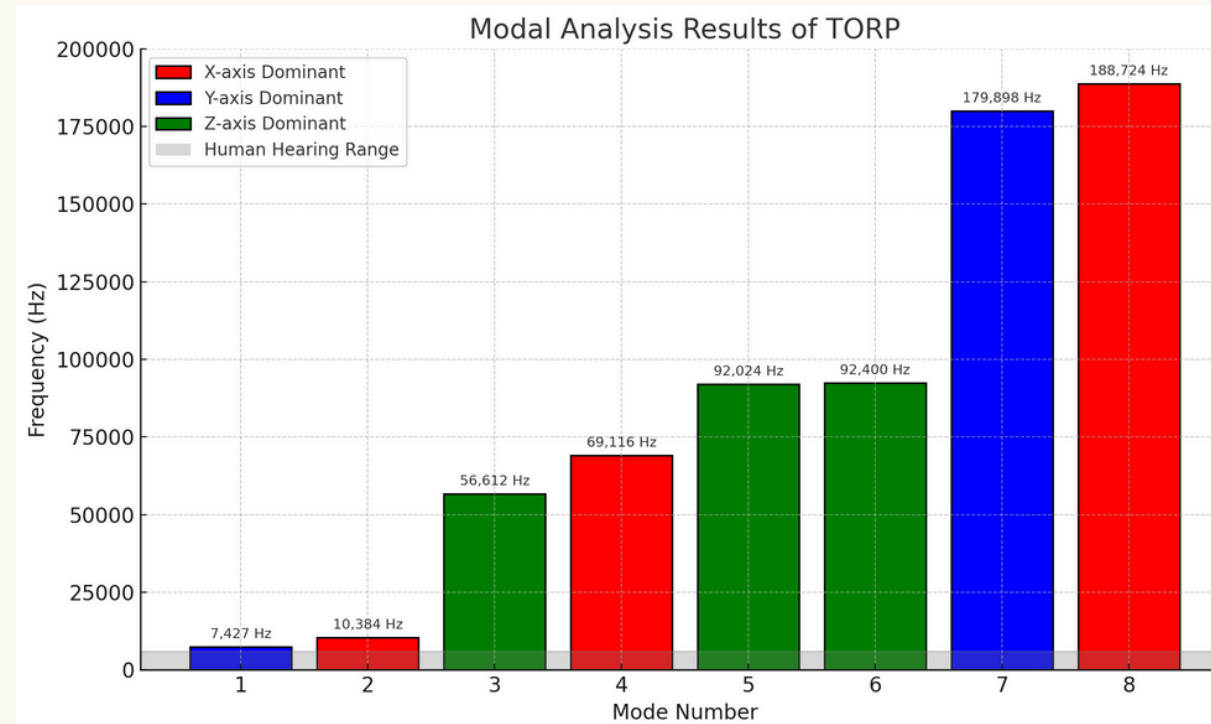
Displacement plot



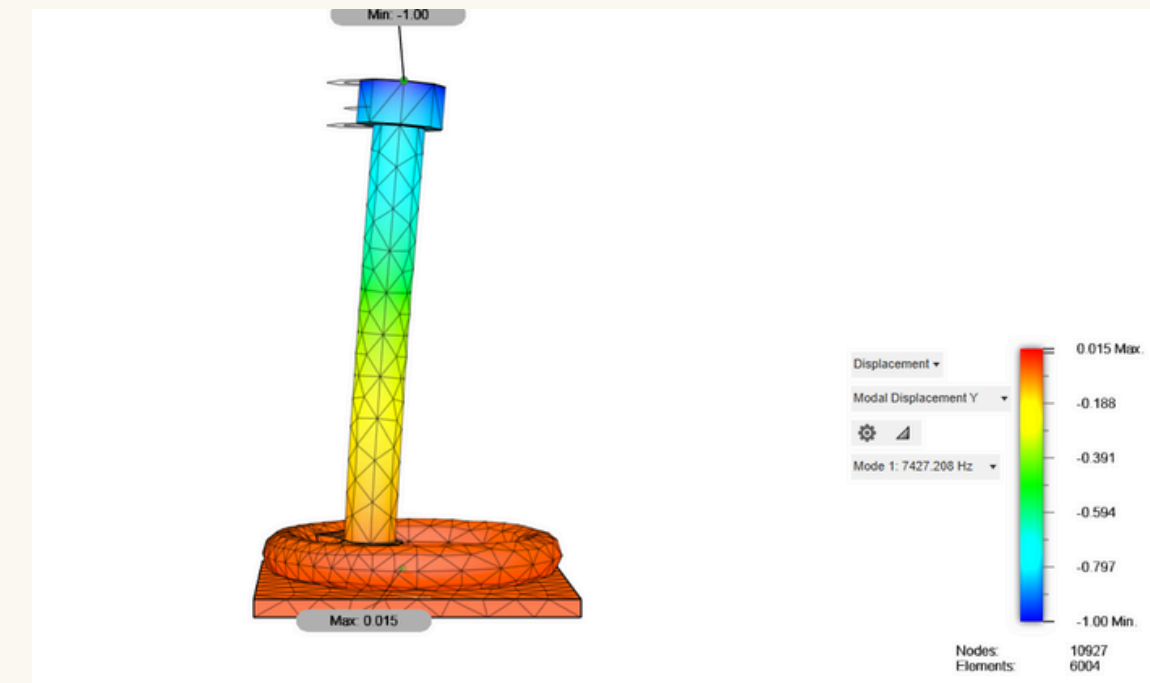
Safety factor plot

Modal Analysis Results

- 1 Natural frequencies: 7.4 kHz – 188.7 kHz
- 2 All modes lie above the 100–6000 Hz hearing band → resonance avoided
- 3 Vibrations are stable.
- 4 Only bending/twisting in higher modes



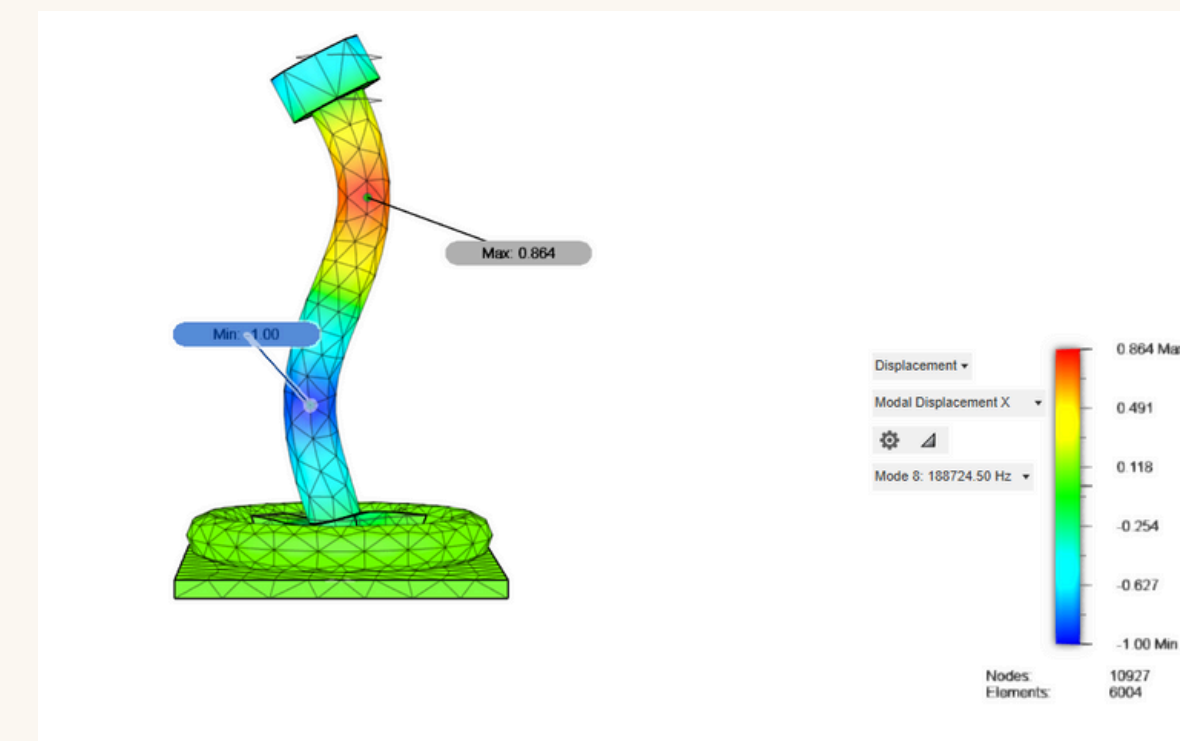
Modal frequencies of TORP – all above hearing range, with bars colored by dominant axis.



Mode 1 deformation shape at 7427 Hz, dominated by Y-axis participation.



Mode 2 deformation shape at 10,384 Hz, dominated by X-axis participation.



Mode 8 deformation shape at 188,724 Hz, dominated by X-axis participation.

Discussion

Static Analysis

- 1** Near-rigid behavior (displacement in nanometres)
- 2** Stress far below yield strength
- 3** High safety factor (15.0)

Modal Analysis

- 1** Natural frequencies = 7.4 kHz to 188.7 kHz
- 2** All above 100–6000 Hz → no resonance
- 3** Stable, well-defined vibration modes

Implication

- 1** TORP is mechanically safe & dynamically reliable
- 2** Provides strong base for prototyping & trials

Limitations & Future Work

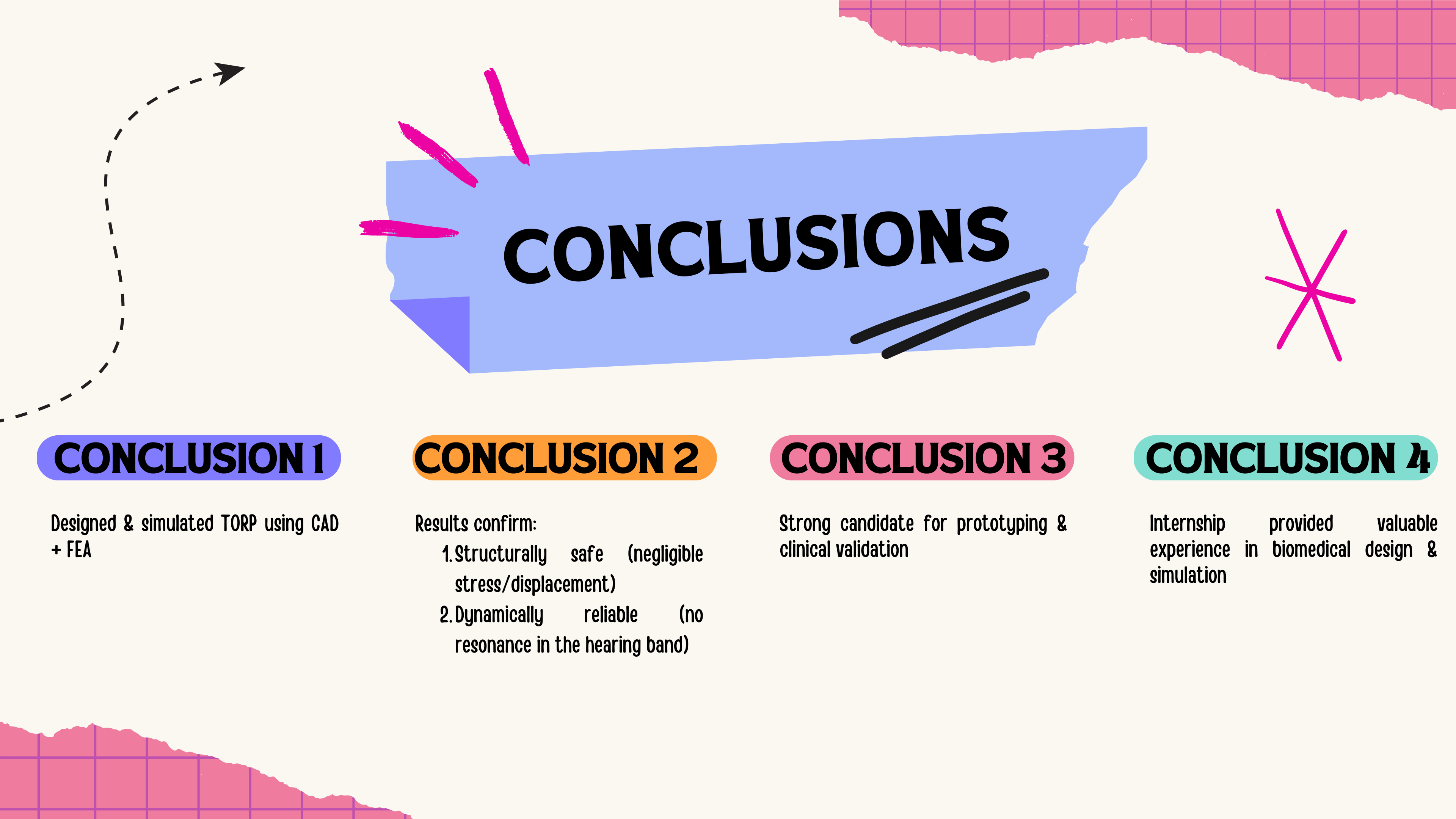
1 LIMITATIONS:

- 1.No tissue interaction/damping modeled
- 2.Fatigue, corrosion, extreme loads not analyzed
- 3.Simplified linear elastic model

2 FUTURE WORK:

- 1.Nonlinear & fatigue analysis
- 2.Prototype fabrication (CNC / 3D printing)
- 3.In-vitro middle-ear testing
- 4.Animal/clinical validation with surgeons





CONCLUSIONS

CONCLUSION 1

Designed & simulated TORP using CAD
+ FEA

CONCLUSION 2

Results confirm:

1. Structurally safe (negligible stress/displacement)
2. Dynamically reliable (no resonance in the hearing band)

CONCLUSION 3

Strong candidate for prototyping & clinical validation

CONCLUSION 4

Internship provided valuable experience in biomedical design & simulation

THANK
YOU!

